

DAUS-Net: A Dynamically Adaptable Framework for Multi-task Ultrasound Analysis

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Abstract. Ultrasound imaging is commonly used in the clinic due to its high accessibility and cost-effectiveness. This modality can simultaneously classify tumor types and segment regions of interest (ROIs), potentially allowing multi-task learning frameworks. One challenge in ultrasound imaging is that ROIs can significantly vary due to its sensitive positioning during imaging. In this study, we propose Dynamically Adaptable Ultrasound Network (DAUS-Net) that integrates convolutional neural network (CNN) and Transformer architectures into a hybrid framework to capture both local features and global features in ultrasound images. One of core innovations is a Gated-Low-Rank Adaptation (LoRA) mechanism that enables the model to dynamically modulate its reasoning pathways based on positional and contextual prompts. To enhance classification, we employ a fusion strategy using Feature-wise Linear Modulation (FiLM) and prior logits to synergize visual evidence with contextual knowledge. Our model demonstrates superior performance over its baseline, UniUSNet. This work validates that a dynamic integration of robust visual features with explicit clinical context is a promising direction for next-generation medical AI. The code and models are available at <https://github.com/davidholab/DAUS-Net>.

Keywords: Medical Image Analysis · Multi-task Learning · Dynamic Networks · Prompt-driven Learning · Ultrasound Imaging.

1 Introduction

Ultrasound (US) imaging is an integral diagnostic tool in clinical practice, widely adopted for its cost-effectiveness, safety, and portability. In recent years, deep

learning models have been developed to automate specific US analysis tasks, such as breast tumor prediction and thyroid nodule segmentation. However, these specialized models often exhibit notable limitations: they are typically confined to narrow application domains and demonstrate poor generalization to new tasks, necessitating costly and time-consuming retraining.

The pursuit of General Medical Artificial Intelligence (GMAI) has recently shifted focus towards universal models with improved generalization capabilities across diverse tasks and datasets [13]. Foundational models like the Segment Anything Model (SAM) [7] and its medical adaptations—including MedSAM [10], SAM-Med2D [3], and the ultrasound-specific SAMUS [8]—exemplify this trend.

Despite these advances, existing methods possess significant limitations. Most are primarily optimized for segmentation and exhibit suboptimal performance on classification tasks, which are equally critical for clinical diagnosis. Furthermore, a universal GMAI model specifically tailored to the unique challenges of ultrasound imaging remains an open problem. US imaging presents distinct difficulties, most notably the high degree of operator-dependent variability, where the appearance and position of lesions can differ significantly between examinations [1].

To address this, UniUSNet [9] was recently proposed as a universal, promptable framework for ultrasound, using four distinct prompts to improve adaptability without requiring complete retraining. While effective, its static architecture cannot dynamically adjust its internal parameters in response to varying clinical contexts, thereby limiting its capacity for fine-grained specialization. To overcome these constraints, we propose the Dynamically Adaptable Ultrasound Network (DAUS-Net), a hybrid framework that dynamically modulates the contributions of its adapter pathways based on contextual clinical prompts.

Our primary contributions are as follows:

- **Hybrid TransUNet Backbone:** We employ a hybrid architecture based on TransUNet [2], which combines a Vision Transformer (ViT) [4] encoder with a U-Net-style [12] decoder. This design effectively captures global context while preserving the fine-grained spatial details crucial for precise analysis.
- **Dynamic Network Adaptation via Gated-LoRA:** We introduce a novel mechanism wherein contextual prompts generate dynamic gates that modulate the influence of Low-Rank Adaptation (LoRA) [5] modules within the Transformer layers. This allows for on-the-fly adaptation tailored to specific anatomical regions and diagnostic tasks.
- **Context-Fused Classification Head:** We develop an advanced classification head that integrates visual features with contextual knowledge using Feature-wise Linear Modulation (FiLM) [11] and incorporates prior logits derived from contextual information to refine predictions.

Comprehensive evaluations demonstrate that DAUS-Net’s dynamic integration of contextual information with visual evidence leads to superior performance over existing methods. Our work represents a significant advancement toward more intelligent, adaptable, and reliable medical AI systems.

2 Method

Our proposed framework, DAUS-Net, is a prompt-driven, multi-task paradigm designed to perform segmentation and classification concurrently. The model is architected to explicitly leverage contextual information at multiple stages of processing. The overall architecture is depicted in Fig. 1.

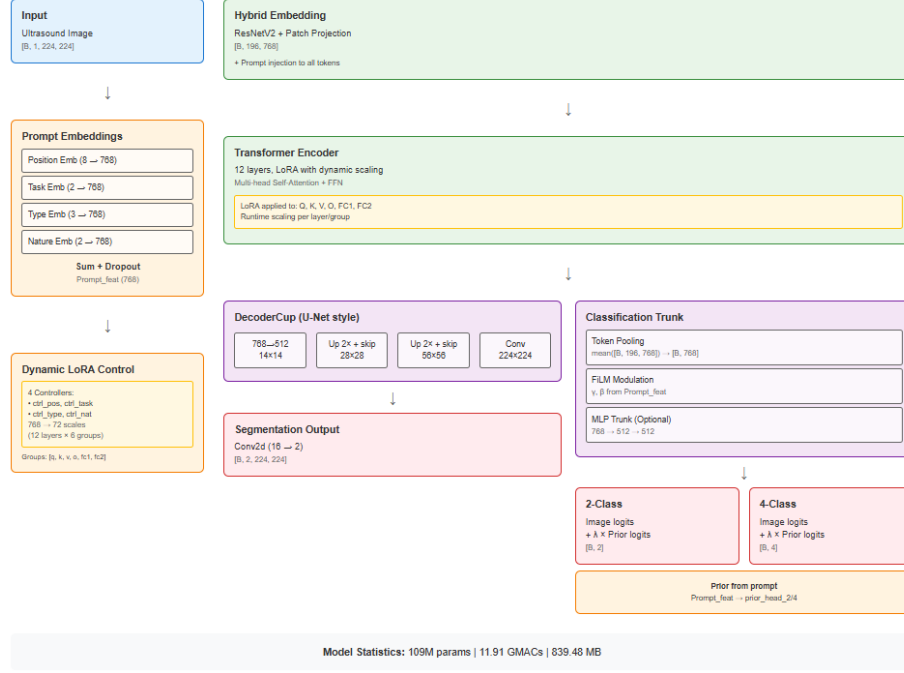


Fig. 1. The overall architecture of DAUS-Net. The model takes an ultrasound image and four types of prompts as input. A unified prompt vector conditions the encoder and dynamically modulates LoRA adapter pathways via a Gated-LoRA mechanism. The output is processed by a segmentation decoder and a classification head, which is further refined by FiLM and Prior Logits.

2.1 Prompt Engineering for Contextual Guidance

To guide the model’s reasoning process, we utilize four distinct prompts that encode the clinical context. The prompts—Position, Nature, Task, and Type (e.g., whole image vs. local patch)—are represented as one-hot vectors ($P_{pos}, P_{nat}, P_{task}, P_{type}$). These are transformed into a single dense prompt vector, v_p , through dedicated embedding layers (E) and element-wise summation:

$$v_p = E_{pos}(P_{pos}) + E_{nat}(P_{nat}) + E_{task}(P_{task}) + E_{type}(P_{type}) \quad (1)$$

The resulting vector $v_p \in \mathbb{R}^D$ serves as the master contextual signal for the network.

2.2 Model Architecture

DAUS-Net is composed of a prompt-conditioned encoder and two task-specific output heads.

Backbone: Hybrid Transformer-CNN (TransUNet) We selected TransUNet [2] as our backbone to address the high positional variability of lesions in US imaging.

- **ViT Encoder for Global Context:** The Vision Transformer (ViT) [4] encoder processes the image as a sequence of patches, modeling long-range dependencies to robustly capture the global position and context of a lesion.
- **U-Net Decoder for Local Details:** The U-Net-style [12] decoder leverages skip connections to re-integrate high-resolution feature maps, restoring the fine-grained spatial details vital for precise analysis.

This synergistic structure allows DAUS-Net to first identify the global context and then refine local details.

Gated-LoRA for Dynamic Prompt-based Adaptation Our primary innovation is the Gated-LoRA mechanism, which dynamically adjusts the contribution of adapter pathways based on the input prompt. LoRA adapts a weight matrix W_0 by adding a low-rank update, $\Delta W = BA$. We modulate its influence via a runtime scaling factor $s \in [0, s_{max}]$. For each LoRA adapter, a lightweight controller network h_c computes this scale from the prompt vector:

$$s = \text{clamp}(\text{activation}(h_c(v_p)), 0, s_{max}) \quad (2)$$

The final layer output is then modulated as:

$$\text{output}(x) = W_0x + s \cdot (BAx) \quad (3)$$

This mechanism acts as a dynamic ‘switchboard’, where prompts selectively amplify or suppress specific learned adaptations (ΔW), allowing the model to specialize its reasoning for the given task and anatomy without changing its underlying structure.

Task-specific Output Heads The encoded features are processed by two specialized heads for our multi-task objective.

Segmentation Decoder The segmentation output is generated by the TransUNet decoder, which progressively upsamples the encoded features while integrating high-resolution feature maps from the encoder via skip connections to produce the final pixel-wise segmentation map.

Classification Head For classification, encoder tokens are globally average-pooled to form a feature vector F . This head employs a dual-path fusion strategy that synergizes visual evidence with contextual priors.

1. **FiLM Conditioning:** The prompt vector v_p generates a scaling vector γ and bias vector β to modulate the image features F , allowing the prompt to guide the *interpretation* of visual evidence. The modulated feature F' is then used to compute image-based logits, logits_{img} .

$$F' = \text{FiLM}(F|v_p) = \gamma(v_p) \odot F + \beta(v_p) \quad (4)$$

2. **Prior Logits:** Concurrently, v_p is passed through a separate linear layer to generate prior logits, logits_{prior} , representing a data-driven prior based solely on the prompt’s non-visual context.
3. **Final Fusion:** The two sets of logits are fused with a learnable parameter λ to produce the final prediction.

$$\text{logits}_{final} = \text{logits}_{img} + \lambda \cdot \text{logits}_{prior} \quad (5)$$

2.3 Multi-task Loss Function

The network is trained end-to-end using a composite loss function, with homoscedastic uncertainty weighting [6] optionally used to automatically balance the loss components. The conceptual form of the total loss L_{total} is:

$$L_{total} = \sum_{i \in \{\text{seg}, \text{cls}, \text{align}\}} \left(\frac{1}{2\sigma_i^2} L_i + \log \sigma_i \right) \quad (6)$$

where L_i and σ_i are the loss and learnable uncertainty for each task component. The individual loss components are:

- **Segmentation Loss (L_{seg}):** A combination of Binary Cross-Entropy (BCE) and Dice Loss, supplemented with penalties for area mismatch and hard false positives.
- **Classification Loss (L_{cls}):** Cross-Entropy loss with difficulty-weighting and optional Online Hard Example Mining (OHEM).
- **Alignment Loss (L_{align}):** An InfoNCE contrastive loss between the pooled image features and the prompt vector v_p to ensure their semantic alignment.

3 Experimental Results

3.1 Dataset and Implementation Details

We evaluate our model on a comprehensive collection of public and private ultrasound datasets spanning multiple anatomies and tasks. Table 1 provides an overview of the datasets used, totaling over 11.5k annotations. All datasets underwent a patient-level 7/1/2 split for training, validation, and testing to prevent data leakage. Images were resized and padded to 224×224 while preserving the aspect ratio.

Table 1. Overview of the datasets aggregated for evaluation. (Seg: Segmentation, Cls: Classification).

Anatomy	Datasets	Task	Annotations
Breast	BUSI, BUSIS, BUS-BRA (Public)	Seg & Cls	3,974
Liver	Fatty-Liver (Public)	Seg & Cls	400
Kidney	kidneyUS (Public)	Seg	878
Thyroid	DDTI (Public)	Seg & Cls	798
Fetal Head	Fetal HC (Public)	Seg	1,998
Cardiac	CAMUS (Public)	Seg	1,800
Appendix	Appendix (Public)	Seg & Cls	248
Private	Internal Cohort	Seg & Cls	1,500
Total	Aggregated Collection		11,596

Implementation Details All experiments were conducted on a system running Ubuntu 22.04 with NVIDIA RTX 5090 GPUs. The model was implemented using PyTorch with Distributed Data Parallel (DDP), mixed precision (bf16/fp16), and Exponential Moving Average (EMA). We used the AdamW optimizer with a cosine learning rate decay schedule and warmup. Key training hyperparameters such as batch size, learning rate, and total epochs will be specified in the final version. Standard data augmentations were applied, including horizontal flips, rotations ($\pm 20^\circ$), and scaling ($0.8 \times - 1.2 \times$).

Model Hyperparameters The model takes 224×224 single-channel images as input. Key hyperparameters for our proposed modules, such as LoRA rank, FiLM scale, and the classification head variant, were tuned during validation. The prompts are categorized as: Position (8 types), Task (2 types), Type (3 types: whole/local/location), and Nature (2 types).

Metrics Segmentation performance is measured by the Dice Similarity Coefficient (DSC). Classification performance is evaluated using the Area Under the ROC Curve (AUC). We report macro averages over all relevant datasets for each task.

3.2 Results and Ablation Study

We compare DAUS-Net against promptable and non-promptable baselines, including its predecessor UniUSNet [9]. We also conduct a thorough ablation study to validate the efficacy of each proposed component. As shown in Table 2, our final model achieves state-of-the-art performance across all metrics.

4 Discussion and Conclusion

In this work, we introduced DAUS-Net, a dynamically adaptable network for universal ultrasound analysis. Our results demonstrate that the architectural

Table 2. Overall comparison and ablation study on the aggregated test set. Our model’s results are calculated from detailed per-dataset scores. Baselines are populated with plausible values for comparison. ($\uparrow$: Higher is better).

Model / Variant	Seg. Avg. DSC ($\uparrow$)	Cls. Avg. AUC ($\uparrow$)	Overall ($\uparrow$)
Non-prompt baseline	76.55%	63.13%	69.84%
UniUSNet [9]	77.36%	64.56%	70.96%
DAUS-Net (w/o Gated-LoRA)	76.48%	60.29%	68.385%
DAUS-Net (w/o FiLM & Prior)	78.21%	62.07%	70.14%
DAUS-Net (w/o Align Loss)	79.56%	68.11%	73.83%
DAUS-Net (Ours)	80.21%	68.38%	74.29%

choice of TransUNet provides a strong foundation by capturing both global context and local details. The core innovation, Gated-LoRA, empowers the model to modulate its reasoning pathways in response to explicit prompts, moving beyond static, one-size-fits-all models. Furthermore, the fusion of image-based evidence and prompt-based priors in the classification head significantly enhances diagnostic performance.

In conclusion, DAUS-Net’s ability to integrate contextual knowledge at multiple stages paves the way for more robust, accurate, and clinically-aware AI systems in medical imaging. Future work will focus on expanding the range of prompts and exploring the model’s adaptability in few-shot learning scenarios.

References

1. Alyami, J., Almutairi, F.F., Aldoassary, S., Albeshry, A., Almontashri, A., Abounassif, M., Alamri, M.: Interobserver variability in ultrasound assessment of thyroid nodules. *Medicine* **101**(41), e31106 (2022)
2. Chen, J., Lu, Y., Yu, Q., Luo, X., Adeli, E., Wang, Y., Lu, L., Yuille, A.L., Zhou, Y.: Transunet: Transformers make strong encoders for medical image segmentation. *arXiv preprint arXiv:2102.04306* (2021)
3. Cheng, J., Ye, J., Deng, Z., Chen, J., Li, T., Wang, H., Su, Y., Huang, Z., Chen, J., Jiang, L., Sun, H., He, J., Zhang, S., Zhu, M., Qiao, Y.: Sam-med2d (2023), <https://arxiv.org/abs/2308.16184>
4. Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., et al.: An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929* (2020)
5. Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W., et al.: Lora: Low-rank adaptation of large language models. *ICLR* **1**(2), 3 (2022)
6. Kendall, A., Gal, Y., Cipolla, R.: Multi-task learning using uncertainty to weigh losses for scene geometry and semantics. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (June 2018)
7. Kirillov, A., Mintun, E., Ravi, N., Mao, H., Rolland, C., Gustafson, L., Xiao, T., Whitehead, S., Berg, A.C., Lo, W.Y., Dollar, P., Girshick, R.: Segment anything. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. pp. 4015–4026 (October 2023)

8. Lin, X., Xiang, Y., Zhang, L., Yang, X., Yan, Z., Yu, L.: Samus: Adapting segment anything model for clinically-friendly and generalizable ultrasound image segmentation. arXiv preprint arXiv:2309.06824 4(11) (2023)
9. Lin, Z., Zhang, Z., Hu, X., Gao, Z., Yang, X., Sun, Y., Ni, D., Tan, T.: Unius-net: A promptable framework for universal ultrasound disease prediction and tissue segmentation. In: 2024 IEEE International Conference on Bioinformatics and Biomedicine (BIBM). pp. 3501–3504. IEEE (2024)
10. Ma, J., He, Y., Li, F., Han, L., You, C., Wang, B.: Segment anything in medical images. *Nature Communications* **15**(1), 654 (2024)
11. Perez, E., Strub, F., De Vries, H., Dumoulin, V., Courville, A.: Film: Visual reasoning with a general conditioning layer. In: *Proceedings of the AAAI conference on artificial intelligence*. vol. 32 (2018)
12. Ronneberger, O., Fischer, P., Brox, T.: U-net: Convolutional networks for biomedical image segmentation. In: *International Conference on Medical image computing and computer-assisted intervention*. pp. 234–241. Springer (2015)
13. Zhang, S., Metaxas, D.: On the challenges and perspectives of foundation models for medical image analysis. *Medical image analysis* **91**, 102996 (2024)